

the working prototype of a perovskite module that can power small electronic devices.

Oxford Photovoltaics aims to have a thin-film perovskite solar cell commercially available by year-end. It also plans to design a perovskites-silicon hybrid to produce tandem cells for rooftop panels. As tandem perovskite-silicon cells absorb a larger fraction of the solar spectrum, these are a superior alternative to conventional rooftop silicon panels.

According to report *Rise of Perovskites*, real commercialisation of perovskite photovoltaics is unlikely to happen until the 2019-21 timeframe. The major obstacle in perovskite commercialisation is stability. The light-sensitive material in perovskite photovoltaic devices dissolves or degrades perovskites within hours in the presence of water vapour and decomposes at high temperature or even by prolonged sun exposure. Methods for painting the material on large surfaces also need improvement.

A Stanford University group reported exceptional stability for new perovskite when an organic component called methyl ammonium was replaced with formamidineum or cesium. When encapsulated to protect against moisture, perovskite cells showed no sign of degradation for six weeks, even when exposed to 85°C temperature and 85 per cent relative humidity—a standard test of durability. Stanford scientists are confident that panels that pass this test will usually not fail due to heat and humidity for over 25 years outside.

Several companies claim to have devised methods for sealing and protecting their cells and modules, but the procedures are proprietary. Scientists must also address the possibility of lead contamination before these cells can be commercialised on a large scale. Proprietary sealing methods may help prevent leakage into the environment. Another strategy under development is replacing lead with metals such as tin.

The inordinate low cost of materials as well as low manufacturing cost has led to setting up of commercial perovskite ventures, especially in the United States, aiming for products that could challenge China's monopoly in the global solar market and help spread manufacturing around the world.

The best is yet to come

The progress of third-generation emerging technologies is unprecedented in photovoltaics history from material development and efficiency advancement perspective. These novel technologies would disrupt the future solar cell market, provided their long-term stability and better efficiency issues are addressed with better understanding of materials and device operation principles.

Prof. Prashant V. Kamat of University of Notre Dame's opines that even if the emerging technologies don't replace silicon cells, these are likely to lead to off-shoot technologies and niche applications. At last, we are on the road to see the day when solar becomes an economically substantive part of our energy mix. **EFY**

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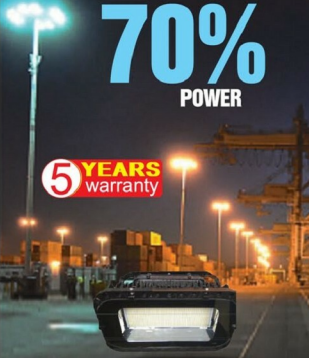
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